

Cover letter for Cancer Genetics

Dear Editors,

Please consider “CSPG4 RNA across sarcoma types with an expanded comparison of extraskeletal myxoid chondrosarcoma” as an original research article for Cancer Genetics.

Extraskeletal myxoid chondrosarcoma is a rare sarcoma for which molecular comparisons are constrained by small specimen collections. This exploratory public-data reanalysis asks where its CSPG4 RNA expression lies among other sarcoma categories. Nine eligible EMC specimens were compared with 393 specimens from 28 other source-defined malignant categories in the same RNA-sequencing collection. The probability-of-superiority score was 0.846 (approximate 95% bootstrap interval 0.709–0.958), with adjusted $P=0.00143$ for the retained 11-gene panel. A sensitivity including all eligible source Intermediate categories gave $A=0.824$ against 489 references. The original low-grade fibromyxoid sarcoma contrast also recurred in a separate array dataset.

The study retains all 644 eligible specimens in a descriptive atlas and adds source-qualified characterization of Treehouse, published historical cohorts, two further GEO collections and OpenPedCan. It documents sample aliases, unresolved diagnoses and measurement differences. GIST, dermatofibrosarcoma protuberans and several benign groups show why high CSPG4 RNA is not unique to EMC. The study makes no claim of diagnostic validation, malignant-cell protein localization or therapeutic benefit.

The work extends an existing public preprint, with historical versions available on Research Square (doi:10.21203/rs.3.rs-10959636/v1) and aiXiv (aixiv.260907.000003). Those public versions precede the present broader comparison. The supplied data, code and supplement distinguish the new analyses from the earlier fixed-panel study. AI assistance with implementation, verification and drafting is disclosed in the manuscript.

The research received no funding. I have no competing financial interests. I disclose that I am an EMC survivor, a non-financial interest relevant to the work. This study reanalyzes public data and includes no newly collected participant specimens or intervention.

Sincerely,

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